Alice Record

CONTACT

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EDUCATION

University of Bath — Bath,UK

Integrated Design Engineering MEng (Hons) — *Sep 2025-May 2030*

- **Year 1:** 2.1
- **YEAR 1 MODULES:** **THERMOFLUIDS** | 73% **SOLID MECHANICS** | 67% **RESPONSIBLE ENGINEERING PRACTICE** | 65%

Moreton Hall

A-levels: Mathematics (A*), Physics (A*), Chemistry (A), Further Mathematics AS (A), EPQ (A*), ESB Grade 8 (Distinction)

ENGINEERING PROJECTS & EXPERIENCE

Team Bath Biodevices — Bath, UK

Design Engineer — *July 2026- Present*

- Designed microfluidic components for a nitrate water-testing device using CAD (Inventor) software
- Optimised the performance of the device by altering the channel's radial distances, length of channels, and corners.
- Formulated and carried out 2 experiments: analysing the influence of heat on resin viscosity and investigating ways to clear resin from channels within the device, reducing clearance time from 30 to 20 minutes.

Smart Insole Personal Project - High Impact Sports — Bath, UK

Project leader | Design Engineer — *June 2026-Present*

- Designing an injury-preventative insole alongside my peers to analyse shock and running gait.
- Managing a team of 3, ensuring that weekly goals are met
- Analysing material characteristics (durability and shock absorption) using laboratory equipment
- Researched, tested and calibrated sensors, microelectronics, and physiology, presenting results in report format.

First year university projects — Bath, UK

Student — *September 2025- May 2026*

- **Ultrasonic sensor project:** Designed a sensor system with 1% percentage error and presented my data (plotted using MATLAB) to my tutor.
- **Ping Pong Ball Launcher:** Worked with a team to produce 5 isometric drawings before finalizing them in Inventor.

Heliotropic solar power system project — Shropshire, UK

Personal Project — *October 2024- March 2025*

- Designed a dual-axis-sensored solar panel for street lights to maximise efficiency in response to the issue of dimmed streetlights across the UK
- Produced a working prototype from LDR's, a TDA2882 IC and resistors in a potential divider circuit.
- Increased energy production by 30%

Sapphiros — Brighton, UK

Work Experience — *August 2024*

- Assessed diagnostic testing devices for Malaria and Covid 19.
- Conducted manufacturing research in flowchart format and presented them to a team of over 15 staff.
- Analysed nano printed circuits for bleeding of metal inks using an Atomic Force Microscope.

COMPUTING SKILLS

3D CAD software: At university I have used Inventor to design and automate an epicyclic gear box and a Ping Pong machine.

Microsoft Office: Intermediate level, using it regularly in studies. I use word to write up reports, Excel to analyse data and powerpoint for presentations

Coding: Proficient skills in Python, Java script and MATLAB. I have used MATLAB in university to present data from labs and code Arduino. I have also coded job scrapers in python and have the following qualifications:

Java Script Certification | 2025
Free Code Academy

Python Certification | 2026
Free Code Academy

SKILLS

Public Speaking & Presentation Skills : *English Speaking Board (presentation on opiate epidemic in America), LAMDA (Distinction- Bronze level), Model United Nations (Chaired and competed in numerous conferences).*

WORK EXPERIENCE

Hazles Farm Childcare, Nursery Assistant & Gardener

July 2025- Present

- *Designed and constructed an interactive sensory and vegetable garden safe for children aged 2-5.*
- *As part of a team we looked after infants from 9 months to 5 years old. This has improved my skills in communication, teamwork and performance under pressure.*
- *Communicated any issues with managerial staff and parents.*

Self Employed Tutor

December 2024- Present

- *Tutored students between yr9-12 in Mathematics, Chemistry and Physics with 100% rate of increased grades.*
- *Communicated with the children and their parents to find gaps in their knowledge*
- *Prepared and organised lessons, producing problem sheets .*

Stem Progression Program for Primary school children

September 2024- December 2024

Volunteer

- *Ran a STEM activities program for local primary school children with the intention to ignite sparks of interest from a young age.*
- *Designed and ran competitions and tasks through the sessions such as producing load be aring bridges*
- *Encouraged girls to get involved*
- *Encouraged children to question and challenge their designs*

INTERESTS

Team Bath Heart (University of Bath)

- *Member of TBH, I help both the design and Mechanical Engineer sectors with activities varying from merch design to CAD (Fusion).*

Running and Hiking

- *Competed in school and County teams for athletics and XC up until ages of 18 and 16 and continue to run recreationally. I also am lucky enough to grow up in Shropshire and enjoy hiking on the weekends.*

Art and Design

- *Embroidery, Woodwork, Pyrography, Jewellery making, Sewing (designing and making my own clothes and accessories, Drawing and Painting. Through this I have had my own businesses through websites and Etsy.*